

# Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

Anonymous WACV Datasets Track submission

Paper ID \*\*\*\*\*

# Beyond Spatial Accuracy: Diagnosing Persistent Orientation Failures in Vision–Language Models

## Supplementary Material

|     |                                                                                               |     |
|-----|-----------------------------------------------------------------------------------------------|-----|
| 001 | 1. Exact Prompts, Decoding, and Parsing                                                       | 045 |
| 002 | VSR plain prompt (identical for 2B and 7B, all conditions; ‘{statement}’ is the VSR caption): | 046 |
| 003 |                                                                                               | 047 |
| 004 | Look at the image carefully.                                                                  | 048 |
| 005 |                                                                                               | 049 |
| 006 | Statement: ”{statement}”                                                                      | 050 |
| 007 |                                                                                               | 051 |
| 008 | Is this statement true or false?                                                              | 052 |
| 009 |                                                                                               | 053 |
| 010 | Answer with exactly one word:                                                                 | 054 |
| 011 | True or False.                                                                                | 055 |
| 012 | 2B structured-decomposition prompt (2B only; single run):                                     | 056 |
| 013 |                                                                                               | 057 |
| 014 | Look carefully at the image.                                                                  |     |
| 015 |                                                                                               |     |
| 016 | Statement:                                                                                    |     |
| 017 | ”{statement}”                                                                                 |     |
| 018 |                                                                                               |     |
| 019 | Determine whether the statement is true                                                       |     |
| 020 | by doing the following:                                                                       |     |
| 021 | 1. Identify the subject object.                                                               |     |
| 022 | 2. Identify the reference object.                                                             |     |
| 023 | 3. Determine the position and                                                                 |     |
| 024 | orientation of the subject relative                                                           |     |
| 025 | to the reference object.                                                                      |     |
| 026 | 4. Pay careful attention to left/right,                                                       |     |
| 027 | front/back, viewpoint, and                                                                    |     |
| 028 | orientation.                                                                                  |     |
| 029 | 5. Decide whether the stated spatial                                                          |     |
| 030 | relationship matches the image.                                                               |     |
| 031 |                                                                                               |     |
| 032 | Answer with exactly one word:                                                                 |     |
| 033 | True or False.                                                                                |     |
| 034 | SITE multiple-choice prompt (replicated from the                                              |     |
| 035 | official lmms-eval task sitebench/utlis.py):                                                  |     |
| 036 | # image examples (one <image> token per visual):                                              |     |
| 037 | <image>...                                                                                    |     |
| 038 | Question: {question}                                                                          |     |
| 039 | Options:                                                                                      |     |
| 040 | A: {opt0}                                                                                     |     |
| 041 | B: {opt1}                                                                                     |     |
| 042 | ...                                                                                           |     |
| 043 | Give me the answer letter directly.                                                           |     |
| 044 | The best answer is:                                                                           |     |
|     | # video examples (pre-prompt):                                                                | 045 |
|     | Select the best answer to the following                                                       | 046 |
|     | multiple-choice question based on the                                                         | 047 |
|     | video. Respond with only the letter                                                           | 048 |
|     | of the correct option.                                                                        | 049 |
|     | Question: {question}                                                                          | 050 |
|     | Options:                                                                                      | 051 |
|     | A: ...                                                                                        | 052 |
|     | B: ...                                                                                        | 053 |
|     | ...                                                                                           | 054 |
|     | Give me the answer letter directly.                                                           | 055 |
|     | The best answer is:                                                                           | 056 |
|     |                                                                                               | 057 |
|     | Decoding. All generations are greedy:                                                         | 058 |
|     | do_sample=False, temperature 0, top- <i>p</i> 1.0.                                            | 059 |
|     | max_new_tokens: 5 for the 2B wrapper, 128 for                                                 | 060 |
|     | 7B VSR runs, 16 for SITE (change validated parse-                                             | 061 |
|     | identical on 125/125 held-out outputs before adoption;                                        | 062 |
|     | recorded in the SITE run metadata).                                                           | 063 |
|     | Parsing. VSR outputs are parsed with an anchored                                              | 064 |
|     | case-insensitive regex (src/evaluation/parser.py):                                            | 065 |
|     | strips prefixes (‘the answer is’, ‘answer:’, ‘response:’,                                     | 066 |
|     | ‘output:’) and trailing punctuation; accepts exact                                            | 067 |
|     | ‘true/yes/correct’ or ‘false/no/incorrect’ forms; sub-                                        | 068 |
|     | string fallback only when exactly one of ‘true’/‘false’                                       | 069 |
|     | appears. Invalid outputs are counted as wrong, never                                          | 070 |
|     | guessed; invalid rate is 0 on all reported runs. SITE                                         | 071 |
|     | outputs use the official parse_multi_choice_response                                          | 072 |
|     | (letter); unparseable responses are counted incorrect                                         | 073 |
|     | and reported.                                                                                 | 074 |
|     | 2. Training and Adapter Configurations                                                        | 075 |
|     | All LoRA conditions share: $r=8$ , $\alpha=16$ , dropout                                      | 076 |
|     | 0.05, AdamW $\text{lr } 10^{-4}$ , weight decay 0.01, linear                                  | 077 |
|     | warmup 10% of steps, 2 epochs, bf16, gradient                                                 | 078 |
|     | checkpointing, gradient clipping 1.0, vision tower                                            | 079 |
|     | frozen. Adapter configs are committed at check-                                               | 080 |
|     | points/*/final/adapter_config.json.                                                           | 081 |
|     | Manifest construction. General: 2,000 state-                                                  | 082 |
|     | ments sampled from the VSR train split. Tar-                                                  | 083 |
|     | getted: orientation-over-sampled subset of the same                                           | 084 |
|     | split. Hard-negative: the orientation block of the                                            | 085 |
|     | general manifest is replaced by audited-clean originals                                       | 086 |
|     | plus paired hard negatives (facing↔facing away from;                                          | 087 |
|     | parallel↔perpendicular) with flipped labels; every in-                                        | 088 |
|     | cluded example was audited for label correctness (451                                         | 089 |

| Condition      | LoRA targets          | Trainable | Batch          | Manifest              |
|----------------|-----------------------|-----------|----------------|-----------------------|
| 2B General     | LM q/k/v/o            | ~2.3M     | microbatch 4×4 | general (2,000)       |
| 2B Targeted    | LM q/k/v/o            | ~2.3M     | same           | targeted (2,000)      |
| 7B LM-only     | LM q/k/v/o            | ~0.3M     | 1              | general (2,000)       |
| 7B HardNeg     | LM q/k/v/o            | ~0.3M     | 1              | hard-negative (2,000) |
| 7B Projector   | visual.merger.mlp.0/2 | 152K      | 1              | general (2,000)       |
| 7B Vision+Proj | merger + blocks 24–31 | 1.46M     | 1              | general (2,000)       |

Table 1. LoRA conditions and their target modules.

train orientation examples audited: 402 clean, 28 excluded, 21 original-only). 2B runs split 5% of the manifest for validation (seed 42). All manifests derive exclusively from the VSR train split; all evaluations use test only.

### 3. Probe Methodology and Full Tables

Features. Qwen2-VL-7B-Instruct, vision tower frozen. ViT patch embeddings (1280-d) and post-merger features (3584-d), mean-pooled per image (ungrounded) or per grounding box (grounded). Grounding boxes from frozen Qwen2-VL grounding (94.6% of 647 orientation images yield both boxes; failures are mostly genuinely absent objects), rescaled to pixels; region features mean-pooled over patch centers inside each box.

Tasks. T1 facing vs. facing away (test  $n=103$ , majority 64.2%); T2 parallel vs. perpendicular ( $n=34$ , majority 64.7%); T3 4-way ( $n=137$ , majority 49.6%). Training on audited-clean VSR train; 5-fold CV; no test examples in training.

Readouts. Linear logistic regression; MLP (256 hidden, ReLU, Adam, early stopping on CV); patch-vote (per-patch linear probe, majority vote over patches); grounded variants with feature sets visual ([subj, ref, subj-ref, subj-ref]), geometry (12-d normalized box features), and visual+geometry.

Ungrounded results (test acc %; balanced acc in parentheses; Wilson 95% CI; majority baselines: T1 64.2%, T2 64.7%, T3 49.6%):

| T1 facing vs. away ( $n=103$ ) | Test | Bal. acc | 95% CI   |
|--------------------------------|------|----------|----------|
| ViT linear                     | 65.0 | 60.4     | [55, 74] |
| ViT MLP                        | 55.3 | 52.0     | [45, 65] |
| Merger linear                  | 68.9 | 64.5     | [59, 77] |
| Merger MLP                     | 52.4 | 49.2     | [42, 62] |
| ViT patch-vote                 | 62.1 | —        | [53, 71] |
| Merger patch-vote              | 60.2 | —        | [51, 69] |

| T2 parallel vs. perpendicular ( $n=34$ ) | Test | Bal. acc | 95% CI   |
|------------------------------------------|------|----------|----------|
| ViT linear                               | 61.8 | 61.0     | [44, 77] |
| ViT MLP                                  | 67.6 | 69.3     | [50, 82] |
| Merger linear                            | 64.7 | 61.4     | [47, 79] |
| Merger MLP                               | 64.7 | 63.3     | [47, 79] |
| ViT patch-vote                           | 73.5 | —        | [57, 85] |
| Merger patch-vote                        | 67.6 | —        | [51, 81] |

| T3 4-way ( $n=137$ ) | Test | Bal. acc | 95% CI   |
|----------------------|------|----------|----------|
| ViT linear           | 52.6 | 40.5     | [44, 61] |
| ViT MLP              | 52.6 | 42.9     | [44, 61] |
| Merger linear        | 53.3 | 44.0     | [45, 61] |
| Merger MLP           | 47.4 | 43.2     | [39, 56] |
| ViT patch-vote       | 51.8 | —        | [44, 60] |
| Merger patch-vote    | 52.6 | —        | [44, 61] |

Grounded results (CV acc / test acc; majority T1 63.7%, T2 57.0%; visual = [subj, ref, subj-ref, subj-ref]; geometry = 12-d box features; T3 4-way is at chance for every readout, test acc 41.7–52.8% across all 12 cells vs. 49.9% majority, committed JSON):

| Readout       | T1 visual | T1 geom.  | T2 visual | T2 geom.  |
|---------------|-----------|-----------|-----------|-----------|
| ViT linear    | 59.8/61.6 | 61.1/60.6 | 51.2/53.6 | 44.2/50.0 |
| ViT MLP       | 59.5/66.7 | 61.8/62.6 | 45.4/46.4 | 45.4/46.4 |
| Merger linear | 57.2/65.7 | 61.1/60.6 | 51.2/53.6 | 44.2/50.0 |
| Merger MLP    | 56.3/52.5 | 61.1/71.7 | 49.0/53.6 | 50.0/42.9 |

The single above-majority cell (geometry-only merger MLP on T1, 71.7% test) has CV 61.1% below the 63.7% majority — not robust — and derives from the grounding model’s own box statistics, not vision-feature content. In short, object-box conditioning with the model’s own grounding predictions and mean-pooled region features does not recover robust orientation decodability under the tested readouts. The T2 patch-vote result (73.5%) is a suggestive point estimate with a wide CI overlapping the majority baseline ( $n=34$ ).

Clear-subset control (canonical frozen snapshot). The 48-example manual audit identified ambiguous or questionable orientation labels. Re-evaluation on progressively stricter subsets (values % from results/tables/orientation\_clean\_label\_table.md, generated by scripts/clean\_label\_orientation.py): Because auditing changes the evaluated sample, strict-subset values are not a matched comparison against other relation families. The earlier probe-side clear/strict check (pre-freeze) additionally suggested that removing audit-flagged examples does not change the probe conclusion: the best ungrounded probe moved only 68.9% → 71.1% → 71.6%

Table 2. Clean-label orientation robustness (canonical frozen snapshot; values % from results/tables/orientation\_clean\_label\_table.md).

| Condition                | Full (137) | −Questionable (132) | Clear (124) | Strict (107) |
|--------------------------|------------|---------------------|-------------|--------------|
| 2B zero-shot             | 62.8       | 65.2                | 66.9        | 75.7         |
| 2B structured            | 53.3       | 52.3                | 51.6        | 53.3         |
| 2B General LoRA          | 62.0       | 63.6                | 64.5        | 72.9         |
| 2B Targeted LoRA         | 64.2       | 65.9                | 68.5        | 76.6         |
| 7B zero-shot             | 63.5       | 65.2                | 68.5        | 72.0         |
| 7B General LoRA          | 65.7       | 67.4                | 70.2        | 78.5         |
| 7B Targeted LoRA         | 64.2       | 65.9                | 69.4        | 72.0         |
| 7B Hard-Neg LoRA         | 66.4       | 68.2                | 70.2        | 76.6         |
| 7B Projector LoRA        | 64.2       | 65.9                | 69.4        | 73.8         |
| 7B Vision+Projector LoRA | 64.2       | 65.2                | 68.5        | 73.8         |

(full/clear/strict,  $n = 103/97/88$ ) and the generative T1 readouts 64.1/66.0/68.2 (zero-shot) and 68.9/72.2/78.4 (General LoRA); committed in results/probe/clear\_subset\_results.json. The reliability of the single-annotator audit underlying this table is assessed with a second blind annotator in §6 (inter-annotator agreement); the frozen table above is retained unchanged.

#### 4. Complete Statistical Test Battery

All pairwise tests are exact McNemar tests (binomial test on discordant pairs, two-sided). Discordant counts are reported as (condition-loses / condition-gains) relative to the control named in each row. No test was run and withheld; the battery below is complete for this paper.

Principal vs. exploratory comparisons. The battery above is fully enumerated; no test was run and withheld. The principal (confirmatory) comparisons are: zero-shot vs. LM-only consistency (ID 19) and the SITE transfer tests (§E). All other comparisons — vision-side conditions and their per-relation cells, hard-negative training vs. control, the remaining consistency comparisons, and the two-stage agreement rows (IDs 15–18) — are exploratory diagnostics reported with exact unadjusted  $p$ -values. Under a Holm adjustment over the 18 accuracy-valid VSR tests ( $\alpha=0.05$ ), only the zero-shot-vs-LM-only consistency test survives (Holm  $p=0.0018$ , using the conservative  $p<0.0001$  bound); the vision-side overall effects (raw  $p=0.0043$ ,  $p=0.0122$ ; Holm 0.073, 0.195) and all other comparisons do not survive a blanket adjustment and are not load-bearing. For the 3 SITE transfer tests, Holm-adjusted  $p$ -values are 0.012 (primary), 1.0, 1.0.

Single-checkpoint note. Every trained condition is a single checkpoint. All McNemar tests above quantify paired disagreement on the fixed test examples between two conditions; they do not estimate training-run variability, and the single-seed results reported here do not

estimate training-run variability. Adapter-rank claims (e.g., the 1–2pp overall differences among trained 7B conditions) should be read in that light.

Consistency verdict patterns. For each strict family and condition, the committed verdict table (results/consistency\_verdict\_table.json, scripts/consistency\_verdict\_table.py) reports the observed both-True / both-False / complementary counts, the original and flipped True rates, and the verdict rates expected under verdict independence given those base rates:

The zero-shot pattern is dominated by a False-defaulting response bias: expected consistency under verdict independence is 30.2%, below the observed 36.9%, so a simple base-rate null does not explain the facing inconsistency; conversely, training raises consistency while introducing both-True contradictions (0%  $\rightarrow$  24.3% LM-only / 17.5% hard-neg), a disclosed specialization trade-off. Full table for left/right ( $n=245$ ), front/behind ( $n=314$ ), and parallel/perp (soft,  $n=34$ ) in the committed JSON.

Consistency definitions. Strict complements (exactly one truth value): left $\leftrightarrow$ right ( $n=245$ ), in front of $\leftrightarrow$ behind ( $n=314$ ), facing $\leftrightarrow$ facing away ( $n=103$ ). Soft complement (parallel $\leftrightarrow$ perpendicular,  $n=34$ ): both-False is legitimate; only both-True is a contradiction; both-True rates are reported separately. Self-consistency on a strict pair means opposite verdicts.

Wilson CIs. All per-relation and SITE subset accuracies are reported with Wilson 95% confidence intervals (main text and committed JSONs).

#### 5. SITE Protocol, Keyword Rule, and Distributions

Pre-specification. Protocol frozen 2026-08-09 before any evaluation; example IDs committed in results/site/site\_protocol.json; the protocol configuration is recorded in results/site/run\_metadata.json. Evaluation-only: no SITE data used for training.

| ID   | Comparison                 | Subset                | Discord. | $p$       |
|------|----------------------------|-----------------------|----------|-----------|
| 1    | HardNeg vs. Gen.           | overall (2195)        | 52/60    | 0.508     |
| 2    | HardNeg vs. Gen.           | weak families         | 20/26    | 0.461     |
| 3    | Projector vs. ctrl.        | overall               | —        | 0.0043    |
| 4    | V+Proj vs. ctrl.           | overall               | —        | 0.0122    |
| 5    | Projector vs. ctrl.        | orientation           | —        | 0.86      |
| 6    | V+Proj vs. ctrl.           | orientation           | —        | 0.85      |
| 7–14 | Proj./V+Proj vs. ctrl.     | per-relation (4×2)    | —        | 0.25–1.00 |
| 15   | Two-stage A-lin. vs. ctrl. | agreement (127 boxed) | 45/16    | 0.0003    |
| 16   | Two-stage A-MLP vs. ctrl.  | agreement (127 boxed) | 47/14    | <0.0001   |
| 17   | Two-stage B-lin. vs. ctrl. | agreement (127 boxed) | 44/20    | 0.0037    |
| 18   | Two-stage B-MLP vs. ctrl.  | agreement (127 boxed) | 47/18    | 0.0004    |
| 19   | Zero-shot vs. LM-only      | consistency 662       | 117/43   | <0.0001   |
| 20   | HardNeg vs. LM-only        | consistency 662       | 31/41    | 0.29      |
| 21   | Projector vs. LM-only      | consistency 662       | 63/48    | 0.18      |
| 22   | V+Proj vs. LM-only         | consistency 662       | 65/56    | 0.47      |

Table 3. Complete VSR test battery (22 rows). Per-relation cells (IDs 7–14) are enumerated in the committed vision\_side\_comparison.json: projector facing  $p=0.77$ , facing-away  $p=1.00$ , parallel  $p=0.25$ , perpendicular  $p=1.00$ ; vision+proj facing  $p=1.00$ , facing-away  $p=0.58$ , parallel  $p=1.00$ , perpendicular  $p=1.00$ . Rows 15–18 are agreement-based: the committed two-stage pipeline scored agreement with the control’s verdicts (not accuracy against ground truth), so these rows are descriptive and are excluded from the confirmatory set and from the Holm family. SITE transfer tests (3: all/primary/secondary, §E) form a separate small family.

Table 4. Facing-pair verdict patterns by condition ( $n=103$ ); “exp.” = expected under verdict independence given the observed base rates.

| Condition (facing pairs, $n=103$ ) | both-T     | both-F     | comp.      | orig-T | flip-T | exp. both-F | exp. comp. |
|------------------------------------|------------|------------|------------|--------|--------|-------------|------------|
| 7B zero-shot                       | 0          | 65 (63.1%) | 38 (36.9%) | 0.20   | 0.17   | 66.5%       | 30.2%      |
| LM-only LoRA                       | 25 (24.3%) | 10 (9.7%)  | 68 (66.0%) | 0.62   | 0.52   | 18.0%       | 49.4%      |
| HardNeg LoRA                       | 18 (17.5%) | 5 (4.9%)   | 80 (77.7%) | 0.60   | 0.52   | 19.0%       | 50.2%      |
| Projector LoRA                     | 18 (17.5%) | 15 (14.6%) | 70 (68.0%) | 0.53   | 0.50   | 23.5%       | 50.3%      |
| Vision+Proj LoRA                   | 14 (13.6%) | 23 (22.3%) | 66 (64.1%) | 0.49   | 0.43   | 27.1%       | 49.4%      |

229 Evaluation: Qwen2-VL-7B- Instruct frozen, greedy,  
 230 official multiple-choice prompt and letter parser,  
 231 max\_new\_tokens=16, images resized to  $\leq 392$ px long  
 232 side (constant protocol parameter; the installed trans-  
 233 formers build ignores max\_pixels), SDPA attention.

234 Subsets. Primary = official category spatial rela-  
 235 tionship reasoning (1,721 examples; 993 images). Sec-  
 236 ondary = heuristic orientation keywords (3,272; 2,024  
 237 images). Exploratory = official movement prediction &  
 238 navigation (1,103; 248 images). Overlaps (image): pri-  
 239 mary  $\cap$  secondary = 488, secondary  $\cap$  exploratory =  
 240 186, primary  $\cap$  exploratory = 0; union = 2,591 images,  
 241 each evaluated once.

242 Exact orientation-keyword rule (applied, case-  
 243 insensitive, word- boundary regex, to question text plus  
 244 non-image option text; src/datasets/site.py):

245 orientation, oriented, orient,  
 246 facing, face, faces, direction,  
 247 directional, view, viewpoint,  
 248 view angle, angle, turned, rotate,  
 249 rotated, rotation, toward, towards,  
 250 away from, left, right, front,

behind, in front of, parallel,  
 perpendicular, clockwise,  
 counterclockwise

251  
252  
253  
254 Secondary subset composition ( $n=2,024$  images) by  
 255 official category: multi-view & cross-image reasoning  
 256 866 (raw 37.8%, CAA 8.8%); spatial relationship rea-  
 257 soning 488 (71.3%, 53.5%); object localization & po-  
 258 sitioning 346 (46.5%, 28.3%); movement prediction &  
 259 navigation 186 (49.5%, 24.3%); 3D information under-  
 260 standing 91 (27.5%, 0.4%); counting & existence 47  
 261 (53.2%, 37.1%). Top sources: exoego4d 316 (36.4%),  
 262 egoexo4d 288 (34.7%), MMTBench 237 (48.9%), SAT  
 263 183 (46.4%), MMIU 133 (28.6%), CLEVR 106 (92.5%),  
 264 SPEC 102 (50.0%), SpatialEval 100 (53.0%), Three-  
 265 DSRBench 95 (61.1%), MMERealWorld 89 (32.6%),  
 266 SeedBench 63 (58.7%), GQA 41 (75.6%), VStarBench  
 267 37, Blink 35, plus 27 smaller sources.

268 Decision rule (pre-specified). (1) If the pri-  
 269 mary subset shows the same broad weakness: run  
 270 the VSR-trained 7B General LoRA on SITE (pre-  
 271 specified step 2). (2) Else if only the orientation  
 272 subset shows the effect: narrow the claim to ob-



ject/direction orientation. (3) Else: report benchmark-specificity. Outcome: branch 2; the claim was narrowed accordingly. The VSR-trained 7B General LoRA adapter was subsequently evaluated on the identical SITE examples as a post-protocol cross-dataset transfer check (no SITE training; predictions in results/site/vsr\_lora\_predictions.csv, paired statistics from scripts/compare\_site\_zeroshot\_lora.py, canonical table in results/tables/site\_validation\_table.md):

Entries are raw accuracy / CAA (%). The adapter is statistically neutral overall, does not improve the orientation subset, and significantly degrades the official spatial-relationship category — VSR gains transfer poorly and can induce negative transfer.

Composition-confound analysis (CPU-only, zero-shot predictions). Because the orientation slice is keyword-derived and heterogeneous, we test whether the heuristic flag predicts lower correctness after controlling for task composition (logistic regression over the 2,591 zero-shot predictions; scripts/site\_confound\_analysis.py → results/site/orientation\_confound\_analysis.json):

Within the official spatial relationship reasoning category, orientation-flagged questions remain descriptively 7.5pp harder than unflagged ones (71.3% vs. 78.8%;  $n=488/505$ ). No source dataset has  $n \geq 30$  in both arms (orientation-heavy corpora are almost entirely flagged, others almost never), so source indicators absorb most of the orientation variance in the full model. Reading: the adjusted association is in the expected direction but not significant; the low aggregate heuristic score is largely explained by task composition. We therefore do not treat SITE as independent confirmation of the VSR orientation bottleneck, and the main text reports the slice as exploratory.

High-precision post-hoc subset (exploratory only). Restricting the heuristic to VSR-construct terms only (facing, facing away, parallel, perpendicular; word-boundary match on question or option text) yields  $n=177$  examples (77 in the official spatial-relationship category, 99 in multi-view/cross-image reasoning, 1 in object localization): raw accuracy 44.1% (95% CI [37.0, 51.4]), CAA -2.3%. This subset is explicitly post-hoc and does not replace the pre-specified frozen definition; it is reported as exploratory only.

## 6. Failure Taxonomy

The 48 persistent-failure statements are the union of: wrong in both 7B zero-shot and 7B LoRA (20); wrong in  $\geq 3$  of 4 base conditions (38); wrong in both 2B and 7B zero-shot (28). A single-annotator exploratory audit (eight-class taxonomy) suggests many persistent errors remain visually interpretable while also identifying

viewpoint and annotation ambiguities; it is qualitative evidence, not a causal localization of the failure. Annotations committed in results/failure\_annotations.csv and results/orientation\_persistent\_annotations.csv.

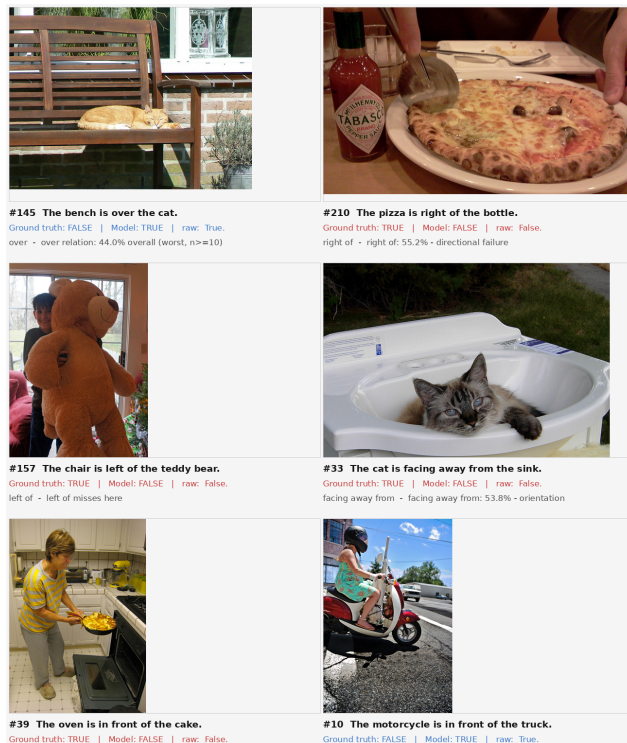


Figure 1. Representative persistent failures (annotated grid; 15 cases). Images: VSR test set. Per the single-annotator exploratory audit, the most frequent class is clear-image reasoning failure: the pose is visually interpretable to a human and the model still errs.

Transition counts. Scaling (2B→7B zero-shot): 28 both wrong, 23 2B-wrong/7B-right, 22 2B-right/7B-wrong, 64 both right. Adaptation (7B zero→7B LoRA): 20 both wrong, 30 fixed, 27 broken, 60 both right. Of the 48 annotated cases, hard-negative training fixes 5; 15 still fail under all three 7B conditions. Facing/facing-away account for 68.8% of persistent failures.

Inter-annotator agreement (additive reliability evidence). Because the clean-label audit (§3) and the taxonomy above are single-annotator, a second, independent human annotator re-rated the same examples under a blind protocol: scripts/export\_iaa\_sheets.py exports two rating sheets (the 137-example orientation test set for the binary clean/ambiguous flag, and the 48 persistent-failure cases for the eight-class taxonomy) containing only image, statement, and relation—ground truth, model predictions, and the first annotator’s labels are withheld—and scripts/iaa\_tool.py

Table 5. SITE VSR-LoRA transfer check (post-protocol; predictions in results/site/vsr\_lora\_predictions.csv, canonical table in results/tables/site\_validation\_table.md). Subsets are defined by the frozen protocol IDs (see note below); no keyword reconstruction is used in analysis.

| Subset                               | Zero-shot   | VSR-LoRA    | $\Delta$ | McNemar $p$ |
|--------------------------------------|-------------|-------------|----------|-------------|
| All images ( $n=2,591$ )             | 54.2 / 31.1 | 53.8 / 30.5 | -0.4     | 0.66        |
| Primary: spatial-rel. ( $n=993$ )    | 75.1 / 59.2 | 71.6 / 53.4 | -3.5     | 0.004       |
| Secondary: orient. kw. ( $n=2,024$ ) | 48.3 / 24.0 | 48.7 / 24.5 | +0.4     | 0.70        |
| Exploratory: movement ( $n=248$ )    | 48.8 / 23.1 | 50.4 / 25.6 | +1.6     | 0.64        |

Table 6. Composition-confound logistic regressions (2,591 zero-shot SITE predictions; outcome = correct).

| Model: correct $\sim$ orient + controls     | OR (orient) | 95% CI       | $p$                   |
|---------------------------------------------|-------------|--------------|-----------------------|
| Unadjusted                                  | 0.31        | [0.25, 0.38] | $2.4 \times 10^{-28}$ |
| + official category                         | 0.74        | [0.57, 0.96] | 0.023                 |
| + category, source, modality, $n_{options}$ | 0.84        | [0.60, 1.16] | 0.29                  |

| Failure mode                          | n  | Share |
|---------------------------------------|----|-------|
| Clear image, model reasoning failure  | 18 | 37.5% |
| Camera viewpoint ambiguity            | 8  | 16.7% |
| Parallel/perpendicular geometry       | 6  | 12.5% |
| Annotation questionable (label wrong) | 5  | 10.4% |
| Intrinsic orientation ambiguous       | 4  | 8.3%  |
| Front/back object ambiguous           | 4  | 8.3%  |
| Small / occluded object               | 2  | 4.2%  |
| Subject/reference inversion           | 1  | 2.1%  |

Table 7. Persistent-failure taxonomy, 48 annotated cases.

provides a blind annotation interface with auto-save. Agreement is computed by scripts/compute\_iaa.py (Cohen’s kappa for the binary flag, Krippendorff’s alpha, nominal, for the taxonomy, both with bootstrap 95% confidence intervals; results/iaa\_results.json).

Results. On the 48 cases rated by both annotators (the persistent-failure set; the remaining 89 orientation examples were never flagged by the first annotator and are excluded from the agreement statistics), the binary clean/ambiguous flag agrees on 32/48 cases (66.7%;  $\kappa=0.36$ , bootstrap 95% CI [0.13, 0.59]). On the eight-class taxonomy the exact-class agreement is 15/48 (31.2%;  $\alpha=0.14$ , CI [-0.03, 0.28]). The disagreement is systematic rather than random: the second annotator never used camera-viewpoint ambiguity or subject/reference inversion, over-used annotation questionable (11 vs. 5) and small/occluded object (11 vs. 2), and flagged 62 of the 137 orientation examples as ambiguous, versus 30 excluded by the first annotator’s strict set. Part of the disagreement on annotation questionable is attributable to the blind protocol itself: judging whether a ground-truth label is wrong requires the label, which was withheld.

Consequences. The frozen first-annotator table (Table 2) is retained unchanged; we do

not redefine the exclusion sets to manufacture agreement. An annotator-dependence analysis (results/iaa/consensus\_subsets.json, scripts/iaa\_consensus\_subsets.py) shows that the sensitivity conclusion is partially annotator-dependent: under the second annotator’s own masks the clean-subset gains largely vanish (7B General LoRA 66.7% and 7B zero-shot 54.7% on  $n=75$ , both similar to or below their full-set values), while under the consensus union mask ( $n=62$ ) the trained conditions again improve (7B General LoRA 77.4%, 7B Hard-Neg 74.2%, 2B zero-shot 72.6%) but 7B zero-shot does not (58.1%). The statement that 2B structured prompting shows no clean-subset gain (53.3% on both full and strict sets) is specific to the frozen first-annotator masks: under the consensus mask it reaches 69.3%. Overall the second annotator provides only partial corroboration, so the clean-label analysis should be read as annotator-specific exploratory evidence rather than a ruling on label noise. No reported accuracy number is altered by this protocol, and no LLM-assisted relabeling was used.

## 7. Reproducibility and Artifact Index

Code: anonymized archive accompanying this submission (no external links). Every headline number in this paper maps to a committed artifact; the audit script scripts/make\_paper\_figures.py recomputes the main tables directly from the raw prediction CSVs (output archived at paper/fig/audit\_output.txt).

Environments. VSR/7B and SITE runs: Ubuntu 24.04, Python 3.12.13, torch 2.12.1+cu130, transformers 5.14.1, datasets 5.0.1, peft 0.20.0, 1 $\times$  RTX A6000 (48 GB). 2B runs: earlier python3.10-era environment (SmolVLM2 wrapper). No lock files exist; the adapter configs (committed) pin the LoRA hyperparameters.

| Headline number                                | Artifact                         | Commit |
|------------------------------------------------|----------------------------------|--------|
| 74.0 / 76.6 / 76.5 / 68.3 (2B)                 | smolvlm2_*_metrics_*.json        | —      |
| 80.9 / 84.7 / 84.3 / 82.9 / 83.1 (7B)          | *_metrics_*.json                 | —      |
| 62.8 / 63.5 / 65.7 / 66.4 (orient. family)     | same + prediction CSVs           | same   |
| Per-relation cells (Table 1 (main))            | orientation_analysis.json        | —      |
| 48-failure taxonomy                            | orientation_persistent_*.json    | —      |
| Probe tables (App. 3)                          | probe/*.json                     | —      |
| Two-stage (App. D, agreement-based rows 15–18) | two_stage_results.json           | —      |
| Consistency (Fig. 4 (main))                    | consistency_stats_all.json       | —      |
| Vision-side p-values                           | vision_side_comparison.json      | —      |
| SITE numbers                                   | site/zeroshot_7b_predictions.csv | —      |
|                                                | site/zeroshot_image_metrics.json | —      |

Table 8. Headline-number → artifact → commit index (all paths relative to results/).

408 Seeds: 42 defaults in 2B paths; single default seed else-  
 409 where — per-run seeds are not recorded in the metrics  
 410 JSONs (a known limitation).

411 Canonical freeze. The experimental result set is a  
 412 frozen snapshot. Canonical tables are generated by  
 413 scripts/build\_canonical\_tables.py; clean-label results  
 414 by scripts/clean\_label\_orientation.py; publication  
 415 figures by scripts/make\_publication\_figures.py;  
 416 SITE paired transfer statistics by  
 417 scripts/compare\_site\_zeroshot\_lora.py;  
 418 VSR/SITE headline recomputation by  
 419 scripts/make\_paper\_figures.py (output archived  
 420 at paper/fig/audit\_output.txt).

421 Two-stage reproducibility caveat. The two-  
 422 stage pipeline requires per-patch embeddings  
 423 (results/probe/patch\_embeddings.pkl, 8.2 GB,  
 424 not committed; regenerable on GPU via  
 425 scripts/extract\_patch\_embeddings.py) and hard-  
 426 codes a Linux working directory. The committed evi-  
 427 dence is the aggregate results/two\_stage\_results.json  
 428 (agreement-based, see App. D); the per-example  
 429 decisions are not committed.

430 SITE secondary-subset audit note (corrected  
 431 before submission). An earlier derived analysis  
 432 (compare\_site\_zeroshot\_lora.py v1, which fed  
 433 the initial zeroshot\_image\_metrics.json-adjacent  
 434 figures and the first canonical site table) re-  
 435 constructed the secondary subset by applying  
 436 ORIENTATION\_KEYWORDS to the question text  
 437 only, yielding  $n=1,824$  (47.3% raw / 22.6% CAA).  
 438 The frozen protocol’s pre-specified definition — re-  
 439 sults/site/site\_protocol.json → frozen\_ids.secondary  
 440 (3,272 examples; 2,024 with images), intersected  
 441 with the 2,591 evaluated IDs — is the authoritative  
 442 subset, and all numbers in this paper use it ( $n=2,024$ ;  
 443 zero-shot 48.3% / 24.0% CAA; VSR-LoRA 48.7% /  
 444 24.5%,  $p=0.70$ ). The derived question-only figure was  
 445 corrected before submission; both definitions support  
 446 the same conclusion (orientation  $\ll$  primary), but only

the frozen-ID set matches the pre-specification. No  
 prediction CSV was modified and no model was rerun.

Data licenses/access. VSR: cam-  
 bridgelt1/vsr\_random (HF). SITE: franky-  
 veteran/SITE-Bench, CC-BY-4.0 (media in 130 GB  
 zips; requires HF read token). LoRA adapters are  
 committed and loadable with peft; base-model weights  
 via HF Hub.

Exact artifacts for the probe tables  
 (App. 3): results/probe/probe\_results.json,  
 results/probe/patch\_probe\_results.json, re-  
 sults/probe/grounded\_probe\_results.json, re-  
 sults/probe/clear\_subset\_results.json, re-  
 sults/probe/grounded\_boxes.json.